Soil moisture analysis via oven drying

Quick Overview

Parameter	Information
Protocol ID	PHYS-002
Version	1
Method Owner	Lieke Lipsch
Estimated Hands-on Time	~3 minutes per sample
Total Duration	2 - 4 days of drying
Difficulty Level	☒ **Basic** ☐ Intermediate ☐ Advanced
Training Required	☐ Yes ☒ **No**
Biosafety Level	/
Typical Throughput	100 (Depending on oven size and sample size)
Sample types	Soil samples
Output	Gravimetric soil moisture
Related Protocols	PHYS-003 Soil moisture analysis via soil moisture analyser

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Method Information

Method Title: Soil moisture analysis via oven drying

Author(s): Lieke Lipsch

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Last Updated: 07-09-2026

Reviewed By: Franziska Raab

Approved By:

Purpose

This method is used to determine the soil moisture of soil samples.

Scope

Suitable for

- Fresh soil samples

Not suitable for

- Old/ dried out soil samples

Safety Considerations

Hazards

- Biological: /
- Chemical: /
- Physical: /

Required PPE

- Lab coat
- Gloves

Waste Disposal

- Biological waste: Contaminated soil should be disposed of appropriately
- Chemical waste: /
- Sharps: /

Materials and Reagents

Reagent/Material	Specification	Supplier	Storage
Aluminium weighing boats	Different diameters available	https://www.sigmaaldrich.com/DE/de/product/sigma/hs14521a?srsltid=AfmBOoqYCGW0WUaCl45kB_BhCzw7R6-Yu-3n4aD0UlkWIijvn2psZGJ_	Usually stored in the drawers underneath the soil moisture analyser in the soil lab (205)

Equipment

Equipment	Model	Location
Drying oven	Different models will work	2x in soil lab 205
Scale	Different models will work	2x in soil lab 205

Procedure

Before Starting

Checklist

☐ Required training completed

☐ Reserve drying oven for the duration of the drying steps/ experiment

☐ Equipment available

☐ Sample IDs assigned

Step-by-Step Procedure

1. Label the weighing boats with your sample labels
2. Weight the empty weighing boat and note down the weight in a table

3. Put your fresh soil sample (1 - 1.5 gram) in the allotted weighing boat and measure the wet weight including the weighing boat and note this down in the table.
4. Place your samples in the boats into a drying oven at 105°C.
5. Check the weight of your samples over the following days. Once the weight has stabilised you can note the values including the boat weight down as dry weight.
6. Calculate the gravimetric soil moisture by using this equation:

$$Gravimetric\ soil\ moisure\ (\%) = \frac{(Wetweight - boat\ weight) - (Dry\ weight - boat\ weight)}{(Dry\ weight - boat\ weight)}\ x\ 100\%$$

Critical Control Points

⚠: Be very careful when moving your samples.

Why it is critical: If any soil is spilled it will ruin your results.

How to avoid mistakes: move carefully and do not move too many samples at once.

Quality Control

QC Parameter	Acceptance Criterion

Troubleshooting

Problem	Possible Cause	Solution
Too little space in the oven	Too large weighing boats	Use smaller weighing boats (diameter ~25mm) or store samples at room temperature until more space is available
Drying of samples takes a very long time	Use of a large sample size	Reduce sample size, approximately one gram should be sufficient

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Expected Outcome

The gravimetric soil moisture usually ranges between 5 and 50%.

Data Recording in Lab Books

Record the following information:

- Operator
- Date
- Sample IDs
- Batch numbers
- Deviations from protocol
- Raw data location

Storage and Stability

Item	Storage Condition	Stability
Soil samples	Samples should be stored in closed vials or bags to prevent excessive drying.	Soil samples should not be stored long before ‘wet weight’ measurement.

References

- SOPs
- Publications
- Instrument manuals

Revision History

Version	Date	Changes	Author
1.0	07-09-2026	Initial version	Lieke Lipsch
	05-10-2026	Addition of comments	Franziska Raab